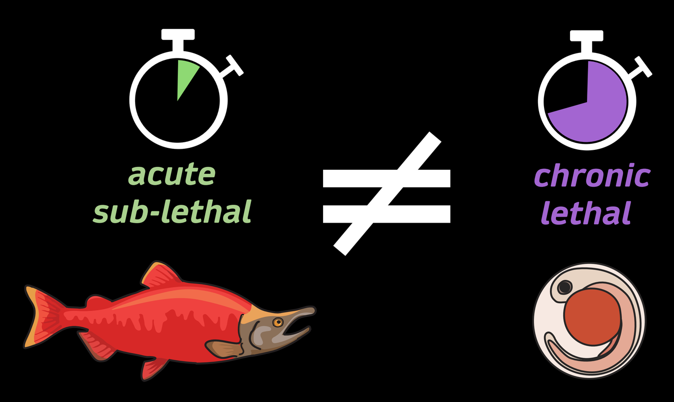
JOIN US!

A distributed experiment on fish embryonic thermal tolerance

THE PROBLEM

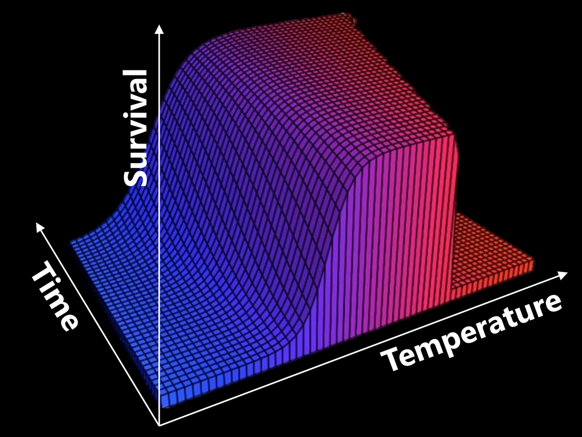
- Fish embryos have been suggested to be **more vulnerable to heat stress than later life stages**
- However, methods are currently **inconsistent** across life stages
- We currently **cannot tease apart biological responses from methodological artifacts**



Endpoints and timing of experiments differ between life stages

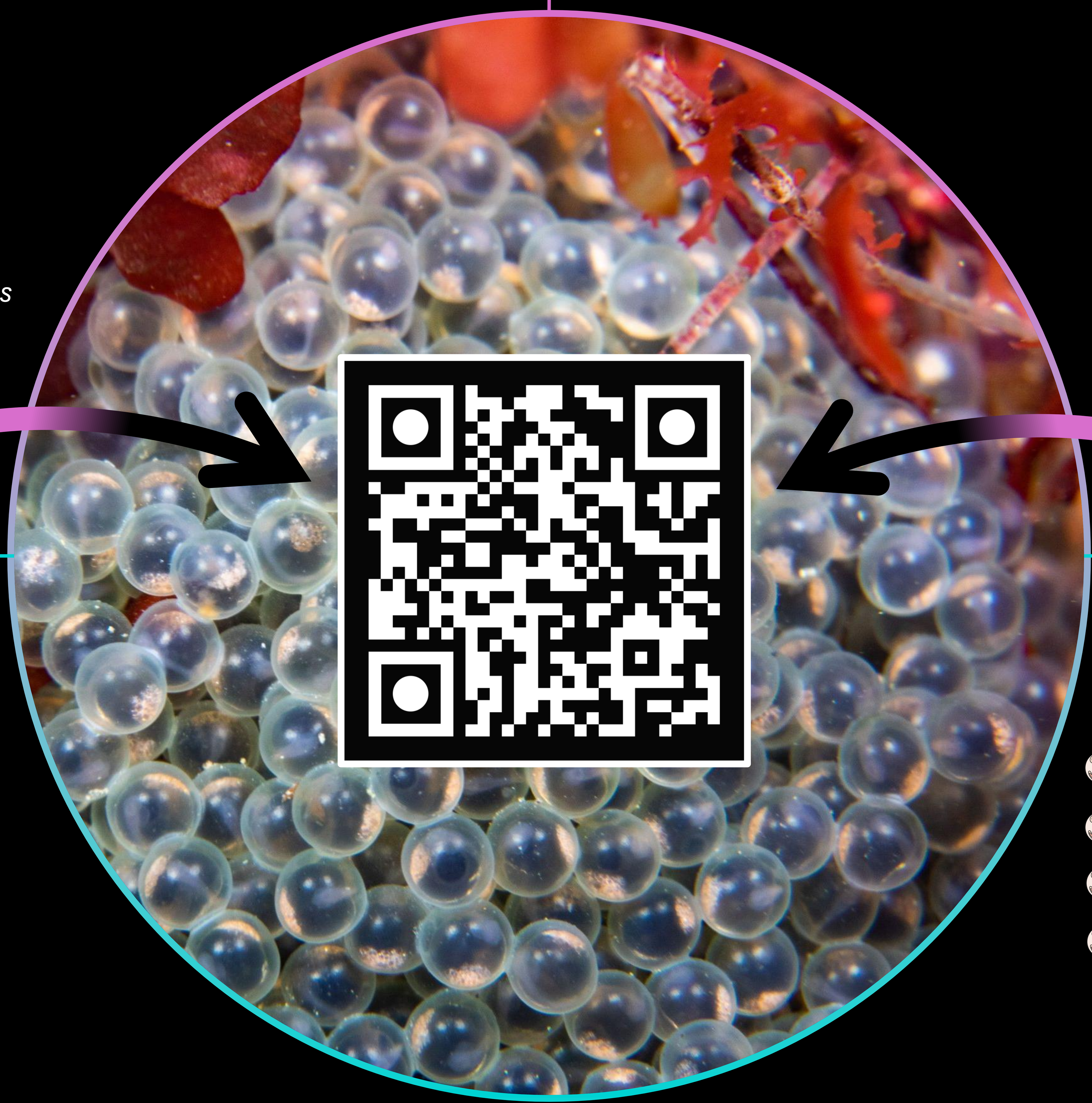
THE SOLUTION

- The thermal tolerance landscape framework can **resolve methodological inconsistencies**
- However, it has not been formally used in fish embryos
- **Globally distributed experiments** allow to test this framework across a large range of species



This framework integrates both the duration and intensity of heat stress

EXPRESS YOUR INTEREST!



EXPRESS YOUR INTEREST!

WHY YOU?

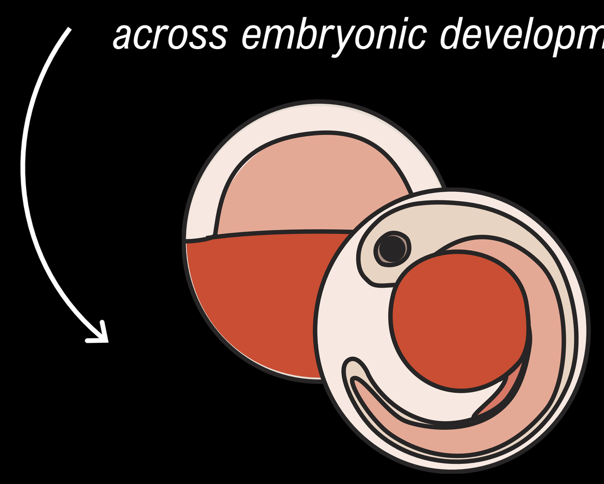
- You are studying **fish**?
- You can collect **eggs**?
- You can control **water temperatures**?
- **We would love your help!**

BENEFITS

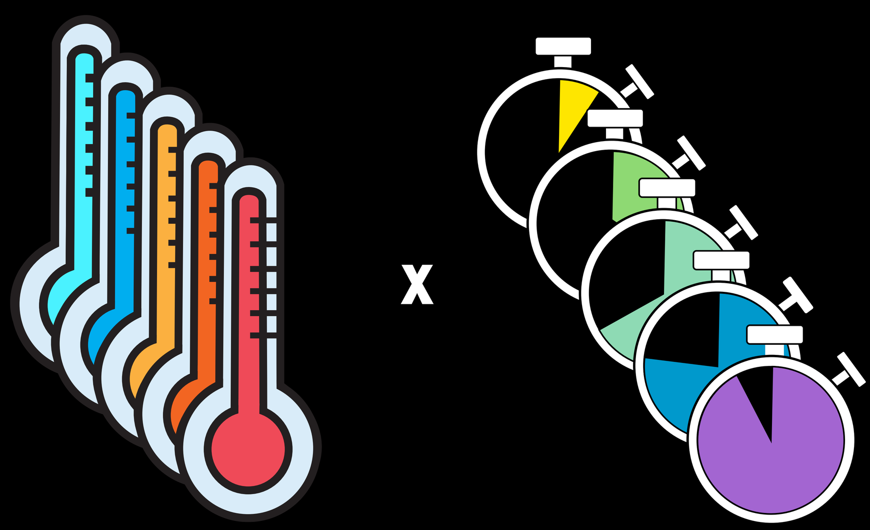
- Insights into an **important biological question**
- **Co-authorship** on the resulting publication
- Access to a **growing collaborative community**
- Regular **workshops, events, and discussions**

HOW TO CONDUCT THE EXPERIMENTS

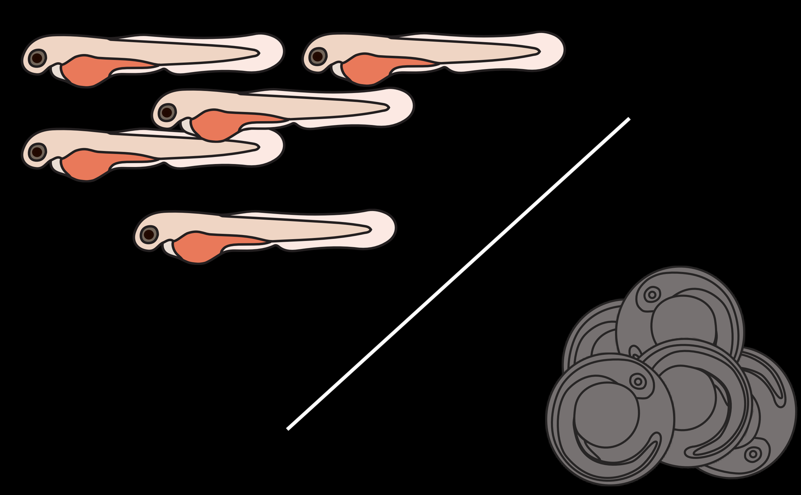
How does thermal tolerance vary across embryonic development?



Early and/or **advanced** embryos



5 temperatures x 5 exposure times



Measure hatching success

SIGNIFICANCE

Identifying **which life stage is most sensitive** to climate warming is **key to predict the impacts of climate change on fish populations**

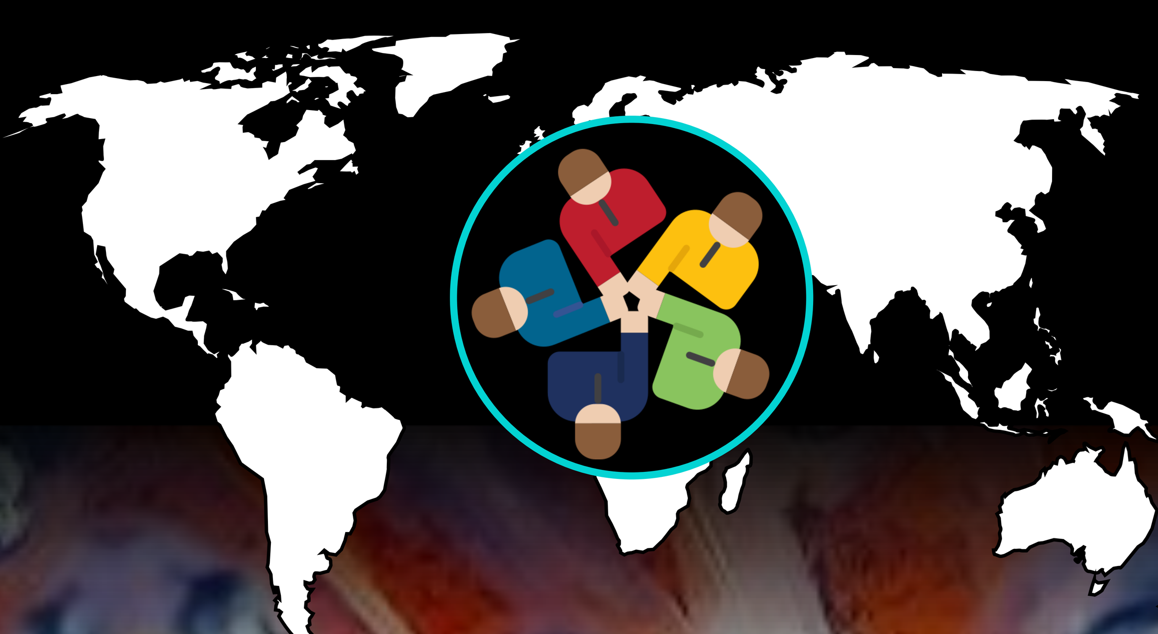
Unifying methodologies will **resolve controversies** over the increased sensitivity of fish embryos relative to other life stages

These results will help us understand **if thermal sensitivity varies throughout the embryonic development**, and **how well embryos can tolerate acute heat stress**

This **globally distributed experiment** will allow the **standardisation of experiments across a large range of fish species and countries**

Globally coordinated efforts pave the way for **more collaborative and efficient research networks** that can help tackle pressing ecological challenges

Let's build a network and tackle this together!



Acknowledgements

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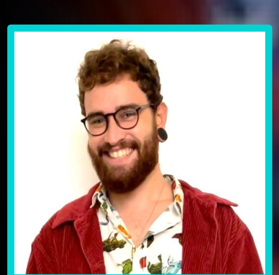

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